ABASYN UNIVERSITY

LAB NO. 03

SYSTEM CALLS

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LAB TASKS

Task 1 (10 marks)

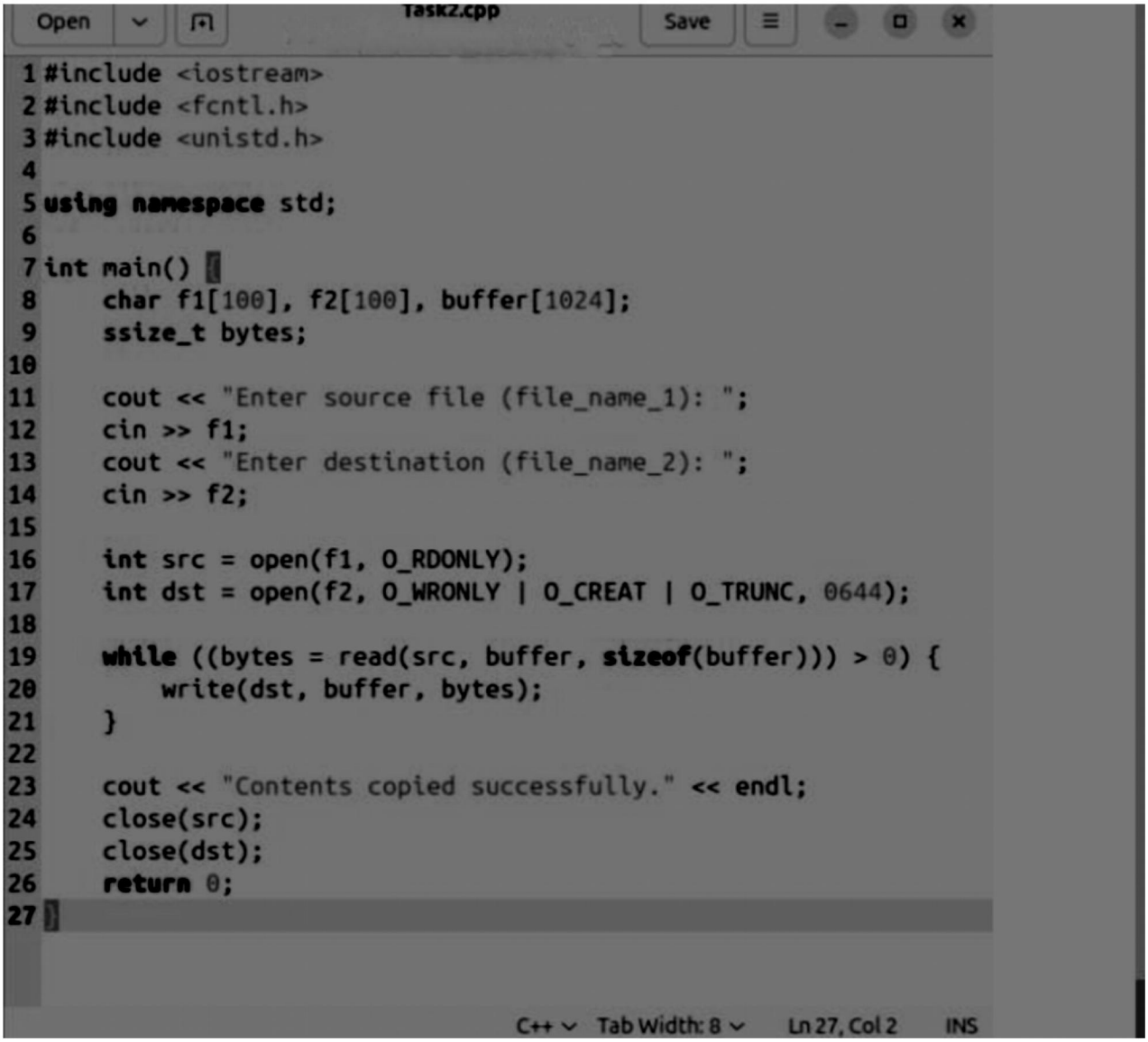
Write a C++ program which takes **file name** from user and creates the file using file management system call. Later on, ask the user to enter the **data** which he/she wants to store in the file and save the file with provided contents. Demonstrate your program using Makefile utility.

Task1.cpp

```cpp
#include <iostream>
#include <fcntl.h>
#include <unistd.h>
#include <string>

using namespace std;

int main() {
    string fileName, content;

    cout << "Enter the file name to create: ";
    getline(cin, fileName);

    int fd = open(fileName.c_str(), O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("Error");
        return 1;
    }

    cout << "Enter data to store: ";
    getline(cin, content);

    write(fd, content.c_str(), content.length());
    cout << "File saved with provided contents." << endl;

    close(fd);
    return 0;
}
```

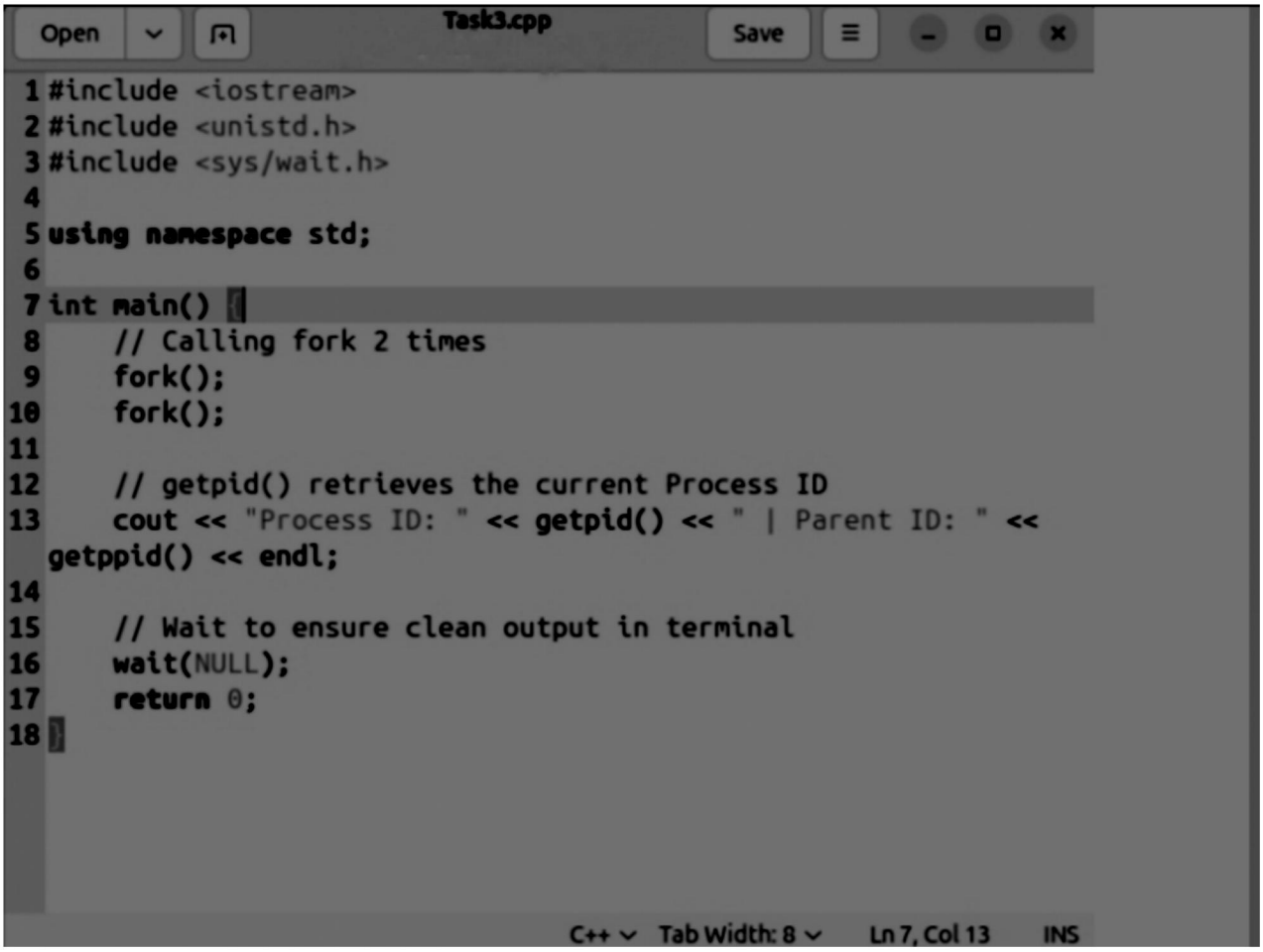
Task 2 (10 marks)

Write a C++ program which takes **names** of **two files**, i.e. *file_name_1* and *file_name_2*. **Copy** the **contents** of the first **file** into the second **file**. Demonstrate your program using Makefile utility.



Task 3 (10 marks)

Write a C++ program which calls **`fork`** function 2 times in main function. Print process ID of each child process using **`getpid`** function. Explore the manual of **`getpid`** function and use it.


Open
Task3.cpp
Save
#include <iostream>
#include <unistd.h>
#include <sys/wait.h>
using namespace std;
int main() {
// Calling fork 2 times
fork();
fork();
// getpid() retrieves the current Process ID
cout << "Process ID: " << getpid() << " | Parent ID: " << getppid() << endl;
// Wait to ensure clean output in terminal
wait(NULL);
return 0;
}
C++
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